Learned Tangent Geometry for Delay-Embedding Anomaly Detection

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Abstract

Unsupervised anomaly detection in a univariate time series asks which windows a healthy system could not have produced. We score a window by its distance from the set of healthy delay windows. Two networks, trained only on the series' own healthy prefix, estimate this distance. A denoiser maps the window towards the healthy set, leaving a residual. A second network, trained by Riemannian Metric Matching, estimates the set's tangent projector, which removes the tangent part of the residual. We show that, for any estimates, the score's error is at most the denoiser's error normal to the set plus the projector's error times the residual's length. The denoiser's pull towards frequently visited regions therefore enters the bound only multiplied by the projector's error. On the 350 held-out series of TSB-AD, evaluated with the benchmark's own protocol, the frozen method reaches a mean VUS-PR of 0.594 (0.594–0.597 over five training seeds). This is not distinguishable from the leading leaderboard entry, a pretrained and fine-tuned foundation model (mean per-series difference $+0.008$, 95% interval $[-0.030, +0.047]$). Our score is also above the nearest-neighbour distance computed with the same windows and readout ($+0.025$, $[+0.004, +0.047]$). The denoiser residual alone is not distinguishable from our score, so the projector adds little on these data. Training both networks costs a median of 1.5 PFLOPs per series.

1. Introduction

A univariate time series records one quantity over time, in settings from web services and medicine to industry and the environment. Unsupervised anomaly detection on such a series asks which parts of the signal the system could not have produced in healthy operation. A stretch of signal recorded during normal operation is available, but examples of anomalies usually are not. The anomalies that occur later may take forms never seen before, from single-point spikes to changes of shape, level or timing. A detector must therefore learn healthy behaviour from the signal alone, and score every new time point against it.

We treat anomaly detection as measuring how far a window lies from the set of windows a healthy system can produce. Write the windows of a signal $x_1, x_2, \ldots$ as $\mathbf{y}_t = (x_t, \ldots, x_{t+L-1}) \in \mathbb{R}^L$, and let $\mathcal{M} \subset \mathbb{R}^L$ be this healthy set. When the signal comes from a deterministic dynamical system and L exceeds twice the dimension of its attractor, $\mathcal{M}$ has the attractor's dimension for generic observation functions (Takens, 1981; Sauer et al., 1991). Analogous results hold for externally and stochastically forced systems, for each realisation of the forcing (Stark et al., 1997; 2003). A healthy signal gives a finite, uneven sample $\{\mathbf{y}_t\}_{t \le n}$ of $\mathcal{M}$, observed with noise. The density q of these samples reflects the history of the signal, not which regions of $\mathcal{M}$ are healthy. We define anomalous windows as those off the healthy set, $\mathcal{A} \subseteq \mathbb{R}^L \setminus \mathcal{M}$. Anomalies of timing or order qualify once L is long enough to contain them. Detection then becomes estimating the distance $d_{\mathcal{M}}(\mathbf{y}) = \min_{\mathbf{x} \in \mathcal{M}} \|\mathbf{y} - \mathbf{x}\|$ from a new window to a set known only through samples.

A score based on distance to the healthy set separates behaviour that is rare from behaviour that is abnormal. Most detectors depend, directly or indirectly, on how often similar windows occur in the healthy signal. Density and likelihood scores do so by design (Breunig et al., 2000; Su et al., 2019), and isolation forests flag windows that few others resemble (Liu et al., 2008). Reconstruction and forecasting networks, including Transformer detectors and pretrained foundation models, minimise average error over the signal and so fit frequent windows best (Audibert et al., 2020; Xu et al., 2022; Goswami et al., 2024). Nearest-neighbour distance (Ramaswamy et al., 2000) measures distance to the samples rather than to the set, and is inflated where samples are sparse. Foundation models trained on synthetic anomalies (Lan et al., 2026) take their notion of abnormal from the generator of those examples. For each of these, a healthy but rarely visited region can be scored like an anomaly. Dis-

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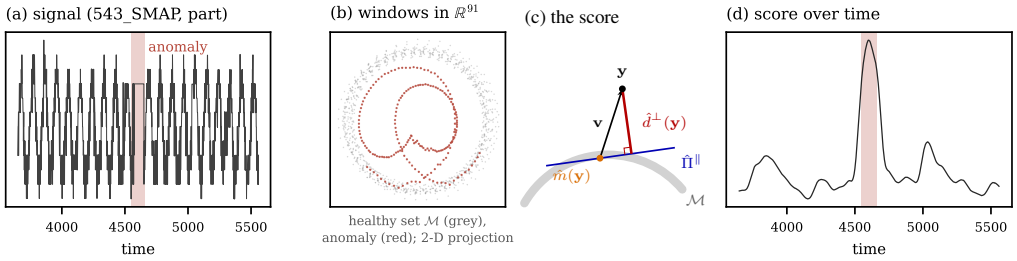


Figure 1. Anomaly detection as distance from a healthy set. (a) Part of series 543_SMAP (TSB-AD development split) with a labelled anomaly (red). (b) Its delay windows ($L = 91$) on their first two principal components: healthy windows (grey) trace a ring, the healthy set $\mathcal{M}$; anomalous windows (red) leave it. (c) Schematic of the score $\hat{d}^{\perp}(\mathbf{y})$, the part of the residual $\mathbf{v} = \mathbf{y} - \hat{m}(\mathbf{y})$ orthogonal to the learned tangent direction $\hat{\Pi}^{\parallel}$. (d) The score over time from the trained networks. This series is an easy case: our score, the denoiser residual, the closed-form CDC score and the nearest-neighbour distance all exceed 0.98 VUS-PR on it.

tance to the set asks instead whether a window belongs, not whether it is common.

We estimate a window's distance from the healthy set with two networks trained only on healthy windows. A denoiser, trained to remove added Gaussian noise, maps a window $\mathbf{y}$ to $\hat{m}(\mathbf{y})$, a Gaussian-weighted average of nearby healthy windows that, at a small noise scale, lies close to the nearest healthy point. The residual $\mathbf{y} - \hat{m}(\mathbf{y})$ has a normal part, which carries the distance to the set, and a tangent part, because the average is pulled towards frequently visited regions. A second network, trained by Riemannian Metric Matching (Bamberger et al., 2026), outputs at each point an $L \times L$ matrix $\hat{\Pi}^{\parallel}$ that approximates the projector $\Pi^{\parallel}$ onto the set's tangent directions. The projector $\Pi^{\parallel}$ is the carré du champ (CDC) operator of the healthy data, evaluated on the coordinate functions. Removing the tangent part of the residual gives the score $\hat{d}^{\perp}(\mathbf{y}) = \|(I - \hat{\Pi}^{\parallel})(\mathbf{y} - \hat{m}(\mathbf{y}))\|$, a learned estimate of the distance to the healthy set. The score needs one forward pass per network and no search over the training windows (Figure 1).

This paper makes three contributions. First, a geometric anomaly score: the distance from a window to the healthy set, estimated by a denoiser and a learned tangent projector (Section 4). Second, an error bound that holds for any estimate, including the trained networks. The score's error is at most the denoiser's error normal to the set plus the projector's error times the residual's length, so the denoiser's tangent pull towards frequently visited regions enters the bound only multiplied by the projector's error (Section 4.1). Both errors can be measured on synthetic data but not on real series; the bound therefore explains the design and shows where the score can fall short, but it does not certify the trained networks. Third, an evaluation on the univariate TSB-AD benchmark (Liu & Paparrizos, 2024), with all choices made on its development split and the frozen method run once on its 350 held-out series (Section 5). There the method reaches a mean VUS-PR of 0.594 (0.594–0.597 over five training seeds). This is not distinguishable from the leading entry of the benchmark's leaderboard (TSB-AD leaderboard, 2026) (mean per-series difference $+0.008$, 95% interval $[-0.030, +0.047]$). That entry fine-tunes a foundation model pretrained on synthetic anomalies (Lan et al., 2026); our networks use no data beyond each series' own healthy prefix. With the same windows, normalisation and readout (our pipeline), our score is above the nearest-neighbour distance ($+0.025$, $[+0.004, +0.047]$) but not distinguishable from the denoiser residual alone. Most of the VUS-PR comes from the denoiser and the pipeline; the projector adds little on these data.

2. Related Work

Time-series anomaly detectors. Most time-series anomaly detectors score how typical a window is under a model fitted to the healthy data. Classical examples include the local outlier factor (Breunig et al., 2000), which compares the density around a window with the density around its neighbours, and isolation forests (Liu et al., 2008), which score how few random splits isolate a window. Deep detectors use likelihoods from mixtures, variational models and flows (Zong et al., 2018; Su et al., 2019; Dai & Chen, 2022) or the error of reconstruction and forecasting networks (Xu et al., 2018; Audibert et al., 2020; Tuli et al., 2022; Wu et al., 2023). Transformer detectors add association and contrastive objectives (Xu et al., 2022; Yang et al., 2023). Time-series foundation models are pretrained on large corpora for forecasting or masked reconstruction (Das et al., 2024; Ansari et al., 2024; Goswami et al., 2024) or, in the case of Time-RCD, on synthetic anomalies (Lan et al., 2026). TSB-AD (Liu & Paparrizos, 2024) compares these families under a common protocol and the VUS-PR metric (Paparrizos et al., 2022), which we follow. In all of these families the score depends on how often patterns occur in the training data. In our score, the denoiser's tangent pull towards frequently sampled regions enters the error bound only multiplied by the projector's error.

Distances between windows. Several detectors measure distances between delay windows directly. Nearest-neighbour distance (Ramaswamy et al., 2000) and its time-series form, the discord of the matrix profile on z-normalised windows (Yeh et al., 2016), score a window by its distance to the nearest training window. Singular-spectrum methods (Moskvina & Zhigljavsky, 2003) and subspace residuals score its distance from a linear subspace of the training windows. Nearest-neighbour distance estimates the distance to the healthy set from samples and is exact only where samples are dense. A single subspace is a flat approximation of a curved set. Normalising each window, as the matrix profile does, discards level and amplitude (Section D). Our networks are designed to estimate the same distance smoothly between samples and to follow the curvature of the set.

Denoising, diffusion and projection. Denoising and diffusion models detect anomalies through reconstruction, inpainting and the estimated noise level (Xiao et al., 2023; Liu et al., 2023; Livernoche et al., 2024), and through the spectrum of the denoiser's posterior covariance (Shoushtari et al., 2026). The inpainting detector of Liu et al. (2023) already scores how far a corrupted and then restored image lands from the original, but it uses many diffusion steps, no tangent projection and no error bound. Geometric readings of generative models explain why likelihood fails for data on manifolds (Kamkari et al., 2024), and topological tests find points that break a data set's geometric structure (Stolz et al., 2020). Closest to ours, Bauer (2026) learns, for images, a correcting map whose fixed points are the normal samples, and scores the residual of this projection. Bauer (2026) argues, as we do, that this avoids flagging rare but normal samples. Its projection changes as few pixels as possible and is learned by inpainting synthetic corruptions. We instead estimate the nearest-point projection by denoising, remove the tangent part of the residual with a learned projector, bound the score's error by two measurable network errors, and work with delay windows of univariate series. These methods share our premise that a model of the healthy data can be read as geometry; we make the geometry explicit.

Learning geometry from samples. Our estimators come from work on learning geometry from samples. Local principal component analysis estimates tangent spaces at a rate set by the sample size and curvature (Aamari & Levrard, 2019). Manifold Parzen windows use the same weighted local covariances (Vincent & Bengio, 2002). Diffusion maps recover the geometry of a point cloud through a random walk, with a density normalisation that separates geometry from sampling (Coifman & Lafon, 2006; Nadler et al., 2006). The carré du champ operator extends this framework and exposes tangent structure (Jones, 2024; Jones & Laners, 2026); Riemannian Metric Matching learns it with a neural network (Bamberger et al., 2026). Denoisers and score functions recover the data manifold at small noise, and their low-noise behaviour encodes the normal directions (Stanczuk et al., 2024). Distributional information enters only at higher order (Shen et al., 2026; Li et al., 2026). We use these results to build a detector and to bound its error.

3. Preliminaries

3.1. Setting

We model healthy windows as noisy samples of a low-dimensional set. A window $\mathbf{y}_t = (x_t, \dots, x_{t+L-1}) \in \mathbb{R}^L$ of a univariate series is written $\mathbf{y} = \mathbf{x} + \boldsymbol{\eta}$ with $\mathbf{x} \sim \mu$, where μ is the distribution of healthy windows, its support $\mathcal{M}$ is the healthy set, and $\boldsymbol{\eta}$ is measurement noise. Embedding theorems make $\mathcal{M}$ low-dimensional for a noise-free deterministic system observed, through a generic observation function, with L above twice its attractor dimension (Takens, 1981; Sauer et al., 1991). We use this as motivation, not guarantee: real series are noisy and often driven by outside inputs, and our L is the period of the signal.

3.2. Geometry of the healthy set

The anomaly score needs the distance to $\mathcal{M}$ and its tangent directions. Where $\mathcal{M}$ is locally a k-dimensional smooth submanifold, let $\Pi^{\parallel}_{\mathbf{x}}$ be the orthogonal projector onto its tangent space $T_{\mathbf{x}}\mathcal{M}$ and $\Pi^{\perp}_{\mathbf{x}} = I - \Pi^{\parallel}_{\mathbf{x}}$ the projector onto the normal space. There we write $\mu = q\,\mathrm{vol}_{\mathcal{M}}$, with $\mathrm{vol}_{\mathcal{M}}$ the k-dimensional volume of $\mathcal{M}$ (arc length for a curve) and q the sampling density: how often healthy windows visit each part of $\mathcal{M}$. The distance and a nearest point $\mathbf{p}^*(\mathbf{y})$ of $\mathcal{M}$ are

$$d_{\mathcal{M}}(\mathbf{y}) = \min_{\mathbf{x}\in\mathcal{M}} \|\mathbf{y}-\mathbf{x}\|, \qquad \mathbf{p}^*(\mathbf{y}) \in \arg\min_{\mathbf{x}\in\mathcal{M}} \|\mathbf{y}-\mathbf{x}\|. \tag{1}$$

The reach ρ of $\mathcal{M}$ is the largest radius within which every point has a unique nearest point (Federer, 1959). Within the reach, the vector from the nearest point to $\mathbf{y}$ has no tangent component. Writing $\mathbf{p}^* = \mathbf{p}^*(\mathbf{y})$,

$$\begin{aligned} d_{\mathcal{M}}(\mathbf{y}) < \rho \implies \Pi^{\parallel}_{\mathbf{p}^*}(\mathbf{y}-\mathbf{p}^*) = 0, \\ d^* := d_{\mathcal{M}}(\mathbf{y}) = \|\mathbf{y}-\mathbf{p}^*\| = \left\|\Pi^{\perp}_{\mathbf{p}^*}(\mathbf{y}-\mathbf{p}^*)\right\|. \end{aligned} \tag{2}$$

The reach is at most the smallest radius of curvature of $\mathcal{M}$ (Aamari et al., 2019, Prop. A.1), and the medial axis, where nearest points are not unique, lies at distance at least ρ from $\mathcal{M}$, with ρ the infimum of that distance. Inside the reach, the distance to the healthy set is the length of the normal part of $\mathbf{y} - \mathbf{p}^*$.

3.3. Geometry from samples: the carré du champ operator

The tangent projector of $\mathcal{M}$ can be read off a diffusion on $\mathcal{M}$, through its carré du champ (CDC) operator. For a

Markov process with generator $\mathcal{L}$, the CDC operator is the bilinear map on functions (Bakry et al., 2014, Def. 1.4.2)

$$\Gamma(f,h) = \tfrac{1}{2}\big(\mathcal{L}(fh) - f\,\mathcal{L}h - h\,\mathcal{L}f\big). \tag{3}$$

For the diffusion whose generator is the Laplace–Beltrami operator Δ of a Riemannian manifold (Brownian motion run at twice the standard speed), $\Gamma(f,h) = g(\nabla f, \nabla h)$. The coordinate function x_i reads the i-th entry of a window. Its gradient in $\mathbb{R}^L$ is the standard basis vector $\mathbf{e}_i$, and its gradient on $\mathcal{M}$ is the tangent part $\Pi^{\parallel}_{\mathbf{x}}\mathbf{e}_i$. Since $\Pi^{\parallel}_{\mathbf{x}}$ is symmetric and idempotent,

$$\Gamma(x_i,x_j)(\mathbf{x}) = \mathbf{e}_i^{\top}\Pi^{\parallel}_{\mathbf{x}}\,\mathbf{e}_j = (\Pi^{\parallel}_{\mathbf{x}})_{ij}, \qquad \Gamma(\mathbf{x}) = \Pi^{\parallel}_{\mathbf{x}}, \tag{4}$$

where $\Gamma(\mathbf{x})$ is the $L \times L$ matrix of these values. The definition needs only a process, not a chart, so it also gives geometry on sets that are not manifolds (Jones, 2024). Applied to the coordinate functions, the CDC operator gives the tangent projector.

A random walk on the healthy windows lets us compute the CDC operator without knowing $\mathcal{M}$, and two networks learn its centred version. From $\mathbf{y}$ the walk jumps to a healthy window $\mathbf{x}$ with Gaussian weights $w_\varepsilon(\mathbf{x} \mid \mathbf{y}) \propto e^{-\|\mathbf{y}-\mathbf{x}\|^2/2\varepsilon}\,\mu(d\mathbf{x})$, the random walk of diffusion maps (Coifman & Lafon, 2006). As $\varepsilon \to 0$ the walk's generator $(P_\varepsilon - I)/\varepsilon$, with P_ε its transition operator, tends to $\frac{1}{2}\Delta$ plus a first-order drift set by the sampling density (Coifman & Lafon, 2006, whose kernel normalisation in footnote 5 gives Δ instead). A drift does not change the CDC operator, so the walk's CDC operator on the coordinate functions tends to $\frac{1}{2}\Pi^{\parallel}$ at points of $\mathcal{M}$; this is the source of the factor 2 below. The networks learn the centred moments (Jones & Lanners, 2026, Eq. 6)

$$m_\varepsilon(\mathbf{y}) = \mathbb{E}_{w_\varepsilon}[\mathbf{x}], \qquad 2\Gamma_\varepsilon(\mathbf{y}) = \tfrac{1}{\varepsilon}\operatorname{Cov}_{w_\varepsilon}[\mathbf{x}], \tag{5}$$

the conditional mean and covariance of a healthy window given its noisy copy $\mathbf{y} = \mathbf{x} + \sqrt{\varepsilon}\,\boldsymbol{\xi}$. A denoiser therefore learns m_ε, and Riemannian Metric Matching learns Γ_ε (Bamberger et al., 2026). The uncentred moment $\frac{1}{\varepsilon}\mathbb{E}_{w_\varepsilon}[(\mathbf{x}-\mathbf{y})(\mathbf{x}-\mathbf{y})^{\top}]$ exceeds $2\Gamma_\varepsilon$ by a rank-one term along the residual $\mathbf{y} - m_\varepsilon(\mathbf{y})$, with eigenvalue $\|\mathbf{y} - m_\varepsilon(\mathbf{y})\|^2/\varepsilon$. Off $\mathcal{M}$ this eigenvalue grows with the squared distance, so the uncentred moment is not close to a projector there; centring removes the term (Section A). The centred moment is also the Jacobian of the barycentre,

$$\frac{\partial m_\varepsilon}{\partial \mathbf{y}}(\mathbf{y}) = 2\Gamma_\varepsilon(\mathbf{y}). \tag{6}$$

Both moments recover the geometry of Section 3.2 at small scale. If $\mathcal{M}$ is compact and $\mathbf{y}$ has a unique nearest point $\mathbf{p}^*(\mathbf{y})$ (in particular within the reach), the weights concentrate at $\mathbf{p}^*(\mathbf{y})$ as $\varepsilon \to 0$ by Laplace's method, which gives the first limit below. If also $2\Gamma_\varepsilon(\mathbf{z}_\varepsilon) \to \Pi^{\parallel}_{\mathbf{p}^*(\mathbf{y})}$ whenever $\mathbf{z}_\varepsilon \to \mathbf{p}^*(\mathbf{y})$ as $\varepsilon \to 0$, which we assume rather than prove (it is plausible for C^2 sets and a positive, continuous density q), the second limit holds:

$$m_\varepsilon(\mathbf{y}) \to \mathbf{p}^*(\mathbf{y}), \qquad 2\Gamma_\varepsilon\big(m_\varepsilon(\mathbf{y})\big) \to \Pi^{\parallel}_{\mathbf{p}^*(\mathbf{y})}. \tag{7}$$

Three departures from the small-scale limits (7) matter in practice. First, they are population limits. With finitely many samples, a very small ε puts almost all the weight on the nearest training window, so m_ε returns that window and $2\Gamma_\varepsilon$ collapses to zero. The scale must therefore stay above the sample spacing. Shen et al. (2026) show that above such a floor, a denoiser-type estimate approximates the nearest point at a rate set by the number of samples. Second, the networks are trained on observed windows, which carry the measurement noise $\boldsymbol{\eta}$. For Gaussian noise $\boldsymbol{\eta} \sim N(0, \sigma^2 I)$, the moments of the observed windows at scale ε are those of the noise-free windows at scale $\varepsilon + \sigma^2$, shrunk towards the identity (Section B). The target's normal eigenvalues rise from about 0 to $\sigma^2/(\varepsilon+\sigma^2)$, so it is close to a projector only when $\sigma^2 \ll \varepsilon$. For homoscedastic noise the score built from these moments is, up to a shift of the point at which the projector is evaluated, a constant multiple of the noise-free score at scale $\varepsilon + \sigma^2$, so the ranking is approximately unchanged. Third, at finite ε the moments average over a neighbourhood of size about $\sqrt{\varepsilon}$. The barycentre m_ε is then pulled in tangent directions towards densely sampled regions, and need not lie exactly on $\mathcal{M}$. The score of Section 4 removes this tangent offset, up to the projector's error, and the error bound of Section 4.1 accounts for the normal offset.

4. Methodology

The method fits two networks to the windows of a healthy prefix and scores every window by a learned estimate of its distance from the healthy set (Algorithm 1). Every choice below is fixed across series.

Windows. Each series $x_1, \ldots, x_n$ comes with a healthy prefix $x_1, \ldots, x_{n_0}$, assumed free of anomalies. We set the window length L to the period of the prefix, estimated by the benchmark's own rule (Liu & Paparrizos, 2024; TSB-AD code, 2026): the lag of the highest peak of the autocorrelation, with a fixed default of 125 when that lag exceeds 300 or no peak exists (Section D). The series is z-normalised with the prefix mean and standard deviation. Each window $\mathbf{y}_t = (x_t, \ldots, x_{t+L-1})$ is divided by $\sqrt{L}$, so that its squared norm is the mean square of its values and does not grow with L. The training set is the $N = n_0 - L + 1$ windows inside the prefix. Training uses only these windows, never labels or test data.

Networks. Both networks are conditioned on the noise scale and trained on noisy pairs: a training window $\mathbf{x}$, a scale ε_{tr}, and $\mathbf{y} = \mathbf{x} + \sqrt{\varepsilon_{tr}}\,\boldsymbol{\xi}$ with $\boldsymbol{\xi} \sim N(0, I_L)$. The denoiser $m_\phi(\mathbf{y}, \varepsilon)$ is trained first, by minimising $\mathbb{E}\,\|m_\phi(\mathbf{y}, \varepsilon_{tr}) - \mathbf{x}\|^2/\varepsilon_{tr}$, whose minimiser is the barycentre m_ε of Section 3.3. It is then frozen. The metric network outputs a factor $M_\theta(\mathbf{y}, \varepsilon) \in \mathbb{R}^{r\times L}$ and sets $\Gamma_\theta = M_\theta^\top M_\theta$, trained with the centred Riemannian Metric Matching loss (Bamberger et al., 2026)

$$\mathbb{E}\Big[\,\|M_\theta M_\theta^\top\|_F^2 - \tfrac{1}{\varepsilon_{tr}}\big\|M_\theta\,(\mathbf{x} - m_\phi(\mathbf{y}, \varepsilon_{tr}))\big\|^2\Big].$$

For $A = M^\top M$ the loss is $\|A\|_F^2 - \operatorname{tr}(AS)/\varepsilon_{tr}$ with $S = \mathbb{E}[(\mathbf{x} - m_\phi)(\mathbf{x} - m_\phi)^\top \mid \mathbf{y}]$, minimised by $S/2\varepsilon_{tr}$ truncated to rank r. With an exact denoiser $S = \operatorname{Cov}[\mathbf{x} \mid \mathbf{y}]$, so $2\Gamma_\theta$ learns the rank-r truncation of the estimate of the tangent projector in Section 3.3; a denoiser error adds a rank-one term (Section A). We use rank $r = 16$, which bounds the tangent dimension the network can represent. Riemannian Metric Matching's low-rank guarantee, proved for the uncentred loss at points of $\mathcal{M}$, requires $r \geq 2k - 1$ for a tangent dimension k (Bamberger et al., 2026, Theorem 4.4), so it covers $k \leq 8$. The scales are drawn as $\log \varepsilon_{tr} \sim N(\log \varepsilon_{eval},\, 1.2^2)$, centred on the scale used for scoring below.

Score. We score at one scale, $\varepsilon_{eval} = 0.64/L$, so that Gaussian noise at this scale has the same total energy, $L\,\varepsilon_{eval} = 0.64$, at every window length. For a window $\mathbf{y}$, let $\hat{m} = m_\phi(\mathbf{y}, \varepsilon_{eval})$ be the denoised point, $\hat{\Pi}^{\parallel} = 2\Gamma_\theta(\hat{m}, \varepsilon_{eval})$ the estimated tangent projector, $\hat{\Pi}^{\perp} = I - \hat{\Pi}^{\parallel}$, and $\mathbf{v} = \mathbf{y} - \hat{m}$. The score is

$$\hat{d}^{\perp}(\mathbf{y}) = \|\hat{\Pi}^{\perp}\,\mathbf{v}\| = \big\|\big(I - \hat{\Pi}^{\parallel}\big)\,\mathbf{v}\big\|,$$

an estimate of the distance $d^* = d_{\mathcal{M}}(\mathbf{y})$ to the healthy set. The projector is evaluated at the denoised point $\hat{m}$, which lies near the healthy set, rather than at $\mathbf{y}$, which may not. The score is small for a healthy window and grows with the window's distance from the set. Scoring needs two forward passes per window and no search over the training windows, so its cost is linear in the length of the series.

Readout. The benchmark scores time points, and each point lies in L windows. Our readout gives time t the Gaussian-weighted mean of the scores of the windows containing it. The weights fall off with the distance between t and each window's centre, with standard deviation $L/4$, so each window contributes mainly to its central points. Averaging the scores of all windows that contain a point is the readout used in a large evaluation of time-series detectors (Schmidl et al., 2022). Weighting towards the window centre follows the aggregation of overlapping patch predictions in image segmentation (Isensee et al., 2021).

Algorithm 1 Learned distance score for one series.

Require: series $x_{1:n}$, healthy prefix length n_0

1: $L \leftarrow$ period estimate of $x_{1:n_0}$; $\varepsilon_{eval} \leftarrow 0.64/L$
2: z-normalise $x_{1:n}$ with prefix statistics; windows $\mathbf{y}_t \leftarrow (x_t, \ldots, x_{t+L-1})/\sqrt{L}$
3: train m_ϕ on prefix windows, then M_θ with m_ϕ frozen, $\log \varepsilon_{tr} \sim N(\log \varepsilon_{eval}, 1.2^2)$
4: **for** each window $\mathbf{y}_t$ **do**
5: $\quad \hat{m} \leftarrow m_\phi(\mathbf{y}_t, \varepsilon_{eval})$; $\mathbf{v} \leftarrow \mathbf{y}_t - \hat{m}$; $\hat{\Pi}^{\parallel} \leftarrow 2M_\theta^\top M_\theta(\hat{m}, \varepsilon_{eval})$
6: $\quad \hat{d}_t^{\perp} \leftarrow \|(I - \hat{\Pi}^{\parallel})\,\mathbf{v}\|$
7: **end for**
8: point scores: Gaussian-weighted mean of $\hat{d}_t^{\perp}$ over windows containing each time point (s.d. $L/4$)

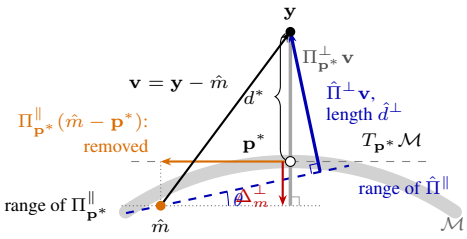


Figure 2. Geometry of Theorem 4.1. A window $\mathbf{y}$ has nearest point $\mathbf{p}^*$ on $\mathcal{M}$ (grey) at distance d^*. The denoised point $\hat{m}$ is offset from $\mathbf{p}^*$ in a tangent direction (orange; the pull towards densely sampled regions), which the projection removes, and normal to it (red; at most $\Delta_m^{\perp}$ over the tube $\mathcal{T}_{\rho'}$). The score $\hat{d}^{\perp}$ is the length of $\hat{\Pi}^{\perp}\mathbf{v}$ (blue), the part of $\mathbf{v} = \mathbf{y} - \hat{m}$ outside the range of $\hat{\Pi}^{\parallel}$, drawn as a line tilted by θ ($\Delta_\Pi = \sin\theta$); it differs from d^* by at most $\Delta_m^{\perp} + \Delta_\Pi\|\mathbf{v}\|$.

4.1. Error bound

The score differs from the true distance to the healthy set by at most the errors of its two networks, and the denoiser's density-driven tangent pull enters this bound only multiplied by the projector's error. Let $\mathbf{y}$ lie within the reach of $\mathcal{M}$, at distance $d^* = d_{\mathcal{M}}(\mathbf{y})$, with nearest point $\mathbf{p}^*$. Assume $\mathcal{M}$ is C^1 near $\mathbf{p}^*$, and let $\Pi^{\parallel}_{\mathbf{p}^*}$ and $\Pi^{\perp}_{\mathbf{p}^*} = I - \Pi^{\parallel}_{\mathbf{p}^*}$ be the tangent and normal projectors there (Section 3.2). Matrix norms are operator norms. The bound holds for any estimates, any point $\hat{m} \in \mathbb{R}^L$ and any matrix $\hat{\Pi}^{\parallel}$, with $\hat{\Pi}^{\perp} = I - \hat{\Pi}^{\parallel}$, $\mathbf{v} = \mathbf{y} - \hat{m}$ and $\hat{d}^{\perp}(\mathbf{y}) = \|\hat{\Pi}^{\perp}\mathbf{v}\|$; the method uses the network outputs of Algorithm 1.

Theorem 4.1 (Error bound)**.** *For any $\hat{m} \in \mathbb{R}^L$ and any matrix $\hat{\Pi}^{\parallel}$,*

$$\big|\hat{d}^{\perp}(\mathbf{y}) - d^*\big| \;\leq\; \|\Pi^{\perp}_{\mathbf{p}^*}(\hat{m} - \mathbf{p}^*)\| + \|\hat{\Pi}^{\parallel} - \Pi^{\parallel}_{\mathbf{p}^*}\|\,\|\mathbf{v}\|. \quad (8)$$

Proof. Adding and subtracting $\mathbf{p}^*$ splits the residual at the nearest point, $\mathbf{v} = \mathbf{y} - \hat{m} = (\mathbf{y} - \mathbf{p}^*) + (\mathbf{p}^* - \hat{m})$; split also $\hat{\Pi}^{\perp} = \Pi^{\perp}_{\mathbf{p}^*} + (\Pi^{\parallel}_{\mathbf{p}^*} - \hat{\Pi}^{\parallel})$. The vector $\mathbf{y} - \mathbf{p}^*$ is normal

(Equation (2)), so $\Pi^{\perp}_{\mathbf{p}^*}(\mathbf{y}-\mathbf{p}^*) = \mathbf{y}-\mathbf{p}^*$ and

$$\hat{\Pi}^{\perp}\mathbf{v} = (\mathbf{y}-\mathbf{p}^*) + \Pi^{\perp}_{\mathbf{p}^*}(\mathbf{p}^*-\hat{m}) + (\Pi^{\parallel}_{\mathbf{p}^*} - \hat{\Pi}^{\parallel})\mathbf{v}.$$

The first term has length d^*, and the triangle inequality gives (8) (Figure 2). □

The bound is deterministic, so it holds for trained networks as well as for ideal ones, and it splits the score's error into two quantities. Both can be measured where $\mathcal{M}$ is known, as on synthetic data; on real series only a lower bound on the projector's error is available. Its first term is the offset of the estimated nearest point from the healthy set in the normal direction; its second is the error of the estimated projector times the length of the residual. The tangential offset $\Pi^{\parallel}_{\mathbf{p}^*}(\hat{m}-\mathbf{p}^*)$, where the sampling density enters the denoised point (Section 3.3), appears only inside $\|\mathbf{v}\|$, multiplied by the projector's error. The denoiser residual instead satisfies $|\|\mathbf{v}\| - d^*| \le \|\hat{m}-\mathbf{p}^*\|$, which contains that offset in full. The score's bound is the tighter of the two when

$$\|\hat{\Pi}^{\parallel} - \Pi^{\parallel}_{\mathbf{p}^*}\|\,\|\mathbf{v}\| < \|\hat{m}-\mathbf{p}^*\| - \|\Pi^{\perp}_{\mathbf{p}^*}(\hat{m}-\mathbf{p}^*)\|.$$

With the exact projector at $\mathbf{p}^*$, the score removes the tangential offset whatever its size. With a large projector error, the score's bound can be looser than the residual's. The method, however, evaluates the projector at $\hat{m}$, not at $\mathbf{p}^*$. For a C^2 set with reach ρ, even the exact tangent projector at a nearest point $\mathbf{p}'$ of $\hat{m}$ on $\mathcal{M}$ differs from $\Pi^{\parallel}_{\mathbf{p}^*}$ by the rotation of the tangent space between the two points: in operator norm, $\|\Pi^{\parallel}_{\mathbf{p}'} - \Pi^{\parallel}_{\mathbf{p}^*}\| \le \|\mathbf{p}'-\mathbf{p}^*\|/\rho$ (Boissonnat et al., 2019, Cor. 3). The tangential offset can therefore also enter through the projector's error. In the ideal setting the projector matters most near $\mathcal{M}$, for healthy windows (Section C).

Theorem 4.1 also tells when an anomalous window is ranked above a healthy one. On the tube $\mathcal{T}_{\rho'} = \{\mathbf{y} : d_{\mathcal{M}}(\mathbf{y}) \le \rho'\}$ with $\rho' < \rho$, let $\Delta^{\perp}_m$, Δ_m and Δ_Π be the largest values of $\|\Pi^{\perp}_{\mathbf{p}^*}(\hat{m}-\mathbf{p}^*)\|$, $\|\hat{m}-\mathbf{p}^*\|$ and $\|\hat{\Pi}^{\parallel} - \Pi^{\parallel}_{\mathbf{p}^*}\|$, respectively, over the windows in $\mathcal{T}_{\rho'}$. Since $\|\mathbf{v}\| \le d^* + \Delta_m$, Theorem 4.1 places every score in a band around its true distance,

$$\left|\hat{d}^{\perp}(\mathbf{y}) - d^*\right| \;\le\; \Delta^{\perp}_m + \Delta_\Pi\,(d^* + \Delta_m),$$

which widens with the distance because the projector error acts on the whole residual. An anomaly is ranked above a healthy window whenever the two bands do not overlap (illustrated in Section C).

Corollary 4.2 (Detection margin). *Let $\mathbf{y}_h, \mathbf{y}_a \in \mathcal{T}_{\rho'}$ be at distances d^*_h and d^*_a from $\mathcal{M}$. If*

$$d^*_a(1-\Delta_\Pi) > d^*_h(1+\Delta_\Pi) + 2\Delta^{\perp}_m + 2\Delta_\Pi\Delta_m,$$

then $\hat{d}^{\perp}(\mathbf{y}_a) > \hat{d}^{\perp}(\mathbf{y}_h)$.

Proof. The condition says that the bottom of the anomaly's band, $d^*_a - \Delta^{\perp}_m - \Delta_\Pi(d^*_a + \Delta_m)$, lies above the top of the healthy window's band, $d^*_h + \Delta^{\perp}_m + \Delta_\Pi(d^*_h + \Delta_m)$. □

Corollary 4.2 guarantees that an anomaly outscores a healthy window once its distance from the healthy set exceeds the healthy window's by a factor $(1+\Delta_\Pi)/(1-\Delta_\Pi)$ plus a margin $(2\Delta^{\perp}_m + 2\Delta_\Pi\Delta_m)/(1-\Delta_\Pi)$. With an exact projector ($\Delta_\Pi = 0$) it only has to be farther by $2\Delta^{\perp}_m$; at $\Delta_\Pi = 0.4$ it has to be more than twice as far. The condition can hold only if $\Delta_\Pi < 1$, since otherwise its left side is not positive; at $\Delta_\Pi = 1$ the estimate $\hat{\Pi}^{\parallel}$ could be the identity, which gives every window a score of zero.

The margin also states what the score cannot do. For the exact distance, an anomaly on the healthy set ($d^*_a = 0$), such as a healthy pattern at the wrong time, scores 0, no more than any healthy window. Corollary 4.2 gives no guarantee for such anomalies; any detection of such anomalies by the trained score comes from estimation error, not from the distance. White measurement noise of variance σ^2 per coordinate places healthy windows near distance $\sigma\sqrt{L-k}$ (for large $L-k$ and $\sigma\sqrt{L-k}$ small compared with the reach), so anomalies must clear that shell by the margin. Anomalies outside the tube are not covered, since $\Delta^{\perp}_m$, Δ_m and Δ_Π are controlled only on $\mathcal{T}_{\rho'}$.

The bound is stated in terms of the network errors, which we do not bound for our networks. Estimators built for the nearest point and the tangent space have known rates in the number of samples (Shen et al., 2026; Aamari & Levrard, 2019), but our networks are not of those types. On real series, Weyl's inequality bounds the learned projector's error from below by the distance of its eigenvalues from those of a projector; its median over windows has median 0.39 over the 48 development series (0.41 to 0.34 on 39 of them after the rescaling of Section C). On these series the guarantee is therefore weak. The bound therefore explains why the score is built as it is and shows where it can fall short, but does not certify the trained networks.

5. Experiments

We test whether the learned distance detects anomalies in real univariate series, and which of its parts matter. The frozen method is run once on the 350 held-out series of TSB-AD (Section 5.2). The 48 development series are used to study the window length and the readout and to locate where the score works and fails (Section 5.3). The held-out run measures the method; the development analysis helps interpret it.

5.1. Experimental setup

Data. We evaluate on the univariate track of TSB-AD (Liu & Paparrizos, 2024), which supplies a development split of 48 series (Tuning) and a held-out split of 350 series (Eva). Every design choice in this paper was made on Tuning or on synthetic data. The method was then frozen and run once on Eva. Each series comes with a healthy prefix assumed free of anomalies. Our method and the five baselines below fit on that prefix only and score the whole series. The series come from 23 source data sets, which we call families. Eva therefore measures the frozen method on data that played no part in its design.

Metric. We report VUS-PR (Paparrizos et al., 2022), the area under precision–recall curves averaged over buffer widths around the labelled anomalies, from zero up to a sliding window. It is the measure chosen by the benchmark's authors, who found it the least sensitive to lags between labels and scores, and unbiased across the cases they compared (Liu & Paparrizos, 2024). VUS-PR depends on the sliding window, so we use the benchmark's own rule: the sliding window is the period estimate of the whole series, as in the official evaluation code. Our window length L uses the same rule on the healthy prefix only. Our sliding windows and VUS-PR match the benchmark's own code (Section D), so our scores are directly comparable with the published ones.

Baselines. Five baselines use our pipeline: the prefix normalisation and, except for the forecast residual, our windows and readout (Section 4). They differ from our method only in how a window or point is scored: (i) **nearest-neighbour distance**, the distance to the closest training window; (ii) **denoiser residual** $\|\mathbf{y}-\hat{m}(\mathbf{y})\|$, our score without the projector; (iii) **PCA residual**, the distance outside the leading principal subspace of the training windows, with the dimension set by a noise threshold on the eigenvalues (Section D); (iv) **closed–form CDC score**, our score with both networks replaced by their closed-form Gaussian-weighted estimates over the training windows; (v) **forecast residual**, the one-step error of a linear autoregression of order L, scored point by point. We also compare with the published Eva result of the leaderboard's leading entry, TimeRCD_MAFT (TSB-AD leaderboard, 2026), a fine-tuned version of the foundation-model detector Time-RCD (Lan et al., 2026). Its result file scores the whole of each series, as we do. Comparing with the pipeline baselines isolates the score; comparing with the published entry measures score and pipeline together.

Statistics. Every comparison scores both methods on the same series. For each series we take the difference in VUS-PR between the two methods, and report the mean of these differences with a bootstrap 95% confidence interval and a Wilcoxon signed-rank p-value. A difference whose interval contains zero is reported as not distinguishable, not as equivalent; where the signed-rank test disagrees with the interval, we say so.

Table 1. Held-out results on the 350 Eva series of TSB-AD (VUS-PR, official sliding window). The difference column is our VUS-PR minus that method's on each series, averaged over the 350 series, with a bootstrap 95% interval; W/L counts the series on which ours is better/worse. Our method is seed 0; with seeds 1–4 each difference moves by at most 0.004. The pipeline baselines share our normalisation and, except the forecast residual, our windows and readout; TimeRCD_MAFT is the leaderboard's published result.

	mean	median	difference [95% CI]	W/L
ours	**0.594**	**0.661**		
same pipeline				
closed-form CDC score	0.589	0.652	$+0.004$ $[-0.016, +0.023]$	189/152
denoiser residual	0.585	0.641	$+0.008$ $[-0.006, +0.022]$	181/160
nearest neighbour	0.568	0.624	$+0.025$ $[+0.004, +0.047]$	205/140
PCA residual	0.497	0.467	$+0.097$ $[+0.054, +0.141]$	212/130
forecast residual	0.362	0.275	$+0.232$ $[+0.188, +0.274]$	271/73
published				
TimeRCD_MAFT	0.586	0.651	$+0.008$ $[-0.030, +0.047]$	179/164

Protocol. The Eva run followed a protocol written before it. The protocol fixed the method, the metric and the statistics. It named the leaderboard's leading entry only as a reference figure. The comparison with that entry and the five pipeline baselines were added after the Eva run, with the baselines' settings fixed on the development split. Nothing was tuned on Eva.

5.2. Main results

Leading entry. On the 350 held-out series the frozen method reaches a mean VUS-PR of 0.594 (median 0.661), against 0.586 (median 0.651) for TimeRCD_MAFT, the leading entry of the benchmark's leaderboard (TSB-AD leaderboard, 2026) (Table 1). The mean per-series difference is $+0.008$, 95% interval $[-0.030, +0.047]$, Wilcoxon $p = 0.39$. The two are not distinguishable, and since the interval allows a deficit of 0.03, equivalence is not shown either. Seed 0 is not a favourable seed. Averaging each series' VUS-PR over the five seeds before taking differences gives $+0.009$, $[-0.031, +0.048]$, against TimeRCD_MAFT.

Pipeline baselines. Most of our method's VUS-PR is also reached by simpler scores computed in the same pipeline (Table 1). Our score is above the nearest-neighbour distance ($+0.025$, $[+0.004, +0.047]$, Wilcoxon $p = 2 \times 10^{-4}$), and above the PCA and forecast residuals. On the development split the nearest-neighbour difference was not distinguishable. By the interval, our score is not distinguishable from the denoiser residual alone ($+0.008$, $[-0.006, +0.022]$), although the signed-rank test gives $p = 0.03$; the projec-

tor's average contribution is small at best. Our score is also not distinguishable from the closed-form CDC score ($+0.004$, $[-0.016, +0.023]$). The closed-form CDC score is a reference, not a ground truth: on synthetic spheres, the learned CDC of Riemannian Metric Matching has 46% lower tangent-space error than a nearest-neighbour estimate for data sets below 4M points (Bamberger et al., 2026, Sec. 5.1). Our networks reach a VUS-PR not distinguishable from it without searching the training windows. The windowing, prefix normalisation and readout are shared by the window-based baselines and carry much of the result.

By family. Similar means hide opposite strengths across families (Table 2). Among families with at least 8 series, our method is ahead of TimeRCD_MAFT by more than 0.1 on MGAB, SVDB, LTDB, UCR and SMD, and behind by more than 0.1 on YAHOO, TODS, Stock and OPPORTUNITY. On development series 560_YAHOO, whose anomaly is a single-point spike, a time-resolved readout (Section 5.3) raises VUS-PR from 0.006 to 1.000. On that series the loss comes from the readout spreading a short anomaly. We have not checked whether this explains the YAHOO family on Eva.

Cost. Training both networks costs a median of 1.5 PFLOPs per series (interquartile range 0.5–4.1; 1.2 EFLOPs for all 350 series), counted from the network sizes and the number of training steps (Section D). At the 19 TFLOP/s measured on RTX PRO 5000/6000 and A6000 GPUs, this is a median of 74 s per series. Scoring costs about 4.6×10^7 FLOPs per window, whatever the number of training windows. The networks have 21–27M parameters, against 37.1M for the released Time-RCD checkpoint (Lan et al., 2026). Time-RCD is pretrained once and reused, whereas ours are trained for each series (Section D).

5.3. Analysis

This section studies the window length and the readout on the 48 development series with the seed-0 networks, and then the method's limitations. Each comparison scores two variants on the same series; the choice of distance is in Section D.

Window length. The best window length follows the length of the anomaly, for our score and for network-free distances alike. To show this, we scored each series at several window lengths and used the labels to find the best one. For our score, the best L, divided by the period, has Spearman correlation $+0.64$, $[+0.37, +0.83]$, with the median labelled anomaly length divided by the period. Spikes are best detected with $L = 8$, anomalies of about a period at the period, and longer anomalies at two to four periods. The nearest-neighbour distance and the PCA residual, which use no networks, show the same pattern (correlations $+0.69$ and $+0.59$). The dependence on L therefore comes from scoring fixed-length windows, not from our networks. The period rule is the best single choice for all three scores. A label-chosen L per series would add $+0.132$, $[+0.079, +0.193]$, to ours, an optimistic ceiling that uses the test labels.

Readout and anomaly length. No single readout suits all anomaly lengths. Our Gaussian readout averages whole-window scores $\|\mathbf{q}_w\|$, where $\mathbf{q}_w = \hat{\Pi}^{\perp}\mathbf{v}_w$ is the normal part of the residual of the window starting at w. A time-resolved readout instead scores time t only by the residual entries that fall on it, $\left(\mathrm{mean}_{w \le t < w+L}\, \mathbf{q}_w[t-w]^2\right)^{1/2}$. It raises VUS-PR on spikes from 0.289 to 0.520 (11 series), but lowers it by 0.13–0.24 on anomalies longer than a tenth of a period. Our readout also raises the VUS-PR of TSB-AD's own nearest-neighbour and PCA detectors on Tuning (Section D). The anomaly length is not known in advance, so we keep the Gaussian readout.

Limitations. The score measures distance from the set of healthy windows learned on the prefix, and its failures follow from that model. A level shift or slow drift moves healthy test windows off that set. Healthy behaviour absent from the prefix is scored as anomalous. On windows far from the training data the metric network's output can be very large (eigenvalues above 10^3 on 3 of 48 series); on those windows the bound of Theorem 4.1 is vacuous. The scoring scale $\varepsilon_{eval} = 0.64/L$ is a fixed rule carried over from synthetic data; a rule set from each series' own noise level was not better on Tuning. Single series can be unstable across training seeds (Section D). These limits, with anomalies that stay on the healthy set (Section 4.1), mark where the method applies.

6. Conclusion

We scored univariate time-series windows by their distance from the set of healthy windows. A denoiser and a learned tangent projector, trained only on each series' healthy prefix, estimate this distance, and we bounded the score's error by the errors of the two networks. On the 350 held-out series of TSB-AD the method is not distinguishable from the leading entry, a fine-tuned foundation model, with no external data and a median of 1.5 PFLOPs of training per series. On these data the projector adds little to the denoiser residual, and much of the result comes from the denoiser and the pipeline. Next steps are a denoiser closer to the healthy set, a readout adapted to the anomaly length, a per-series scale and window length, and multivariate series.

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A. Centring the carré du champ target

Centring removes a rank-one term from the carré du champ target. Fix a query point $\mathbf{y}$ and let $w(\mathbf{x}) \propto e^{-\|\mathbf{y}-\mathbf{x}\|^2/2\varepsilon}$ be the normalised weights the target places on data points $\mathbf{x}$ near $\mathbf{y}$, with weighted centre $m = \int \mathbf{x}\, w(\mathbf{x})\, \mu(d\mathbf{x})$. For any vector $\mathbf{a}$, expanding $\mathbf{x} - \mathbf{a} = (\mathbf{x} - m) + (m - \mathbf{a})$ and using $\int (\mathbf{x} - m)\, w\, d\mu = 0$ gives

$$\int (\mathbf{x}-\mathbf{a})(\mathbf{x}-\mathbf{a})^\top w(\mathbf{x})\, \mu(d\mathbf{x}) = \int (\mathbf{x}-m)(\mathbf{x}-m)^\top w(\mathbf{x})\, \mu(d\mathbf{x}) + (m-\mathbf{a})(m-\mathbf{a})^\top,$$

the parallel-axis identity. With $\mathbf{a} = \mathbf{y}$ (uncentred target) and $\mathbf{a} = m$ (centred target), dividing by ε gives the identity in Section 3.3. The uncentred target therefore adds a rank-one term along $\mathbf{y} - m$, with eigenvalue $\|\mathbf{y} - m\|^2/\varepsilon$, which grows with the distance from the healthy set; the centred target does not. The same identity with $\mathbf{a} = m_\phi$ gives the target of the metric loss when the denoiser is inexact (Section 4). We evaluate the centred moment at m rather than at $\mathbf{y}$. Under the conditions of (7), it tends to the tangent projector at $\mathbf{p}^*(\mathbf{y})$ when evaluated at $m_\varepsilon(\mathbf{y})$. At a point a fixed distance off $\mathcal{M}$, its tangent eigenvalues instead differ from 1 by a curvature-dependent factor.

B. Measurement noise

For Gaussian measurement noise of variance σ^2 per coordinate of the scaled windows, write $\varepsilon_+ = \varepsilon + \sigma^2$, and let m_ε^{obs} and $2\Gamma_\varepsilon^{obs}$ be the moments (5) computed from the observed (noisy) healthy windows. Then

$$\begin{aligned} \mathbf{y} - m_\varepsilon^{obs}(\mathbf{y}) &= \tfrac{\varepsilon}{\varepsilon+\sigma^2}\big(\mathbf{y} - m_{\varepsilon+\sigma^2}(\mathbf{y})\big), \\ 2\Gamma_\varepsilon^{obs}(\mathbf{y}) &= \tfrac{\sigma^2}{\varepsilon+\sigma^2}\, I + \tfrac{\varepsilon}{\varepsilon+\sigma^2}\, 2\Gamma_{\varepsilon+\sigma^2}(\mathbf{y}). \end{aligned} \tag{9}$$

We now derive (9). Let the observed healthy windows be $\mathbf{z} = \mathbf{x} + \boldsymbol{\eta}$ with $\mathbf{x} \sim \mu$ and $\boldsymbol{\eta} \sim N(0, \sigma^2 I)$ independent, and let the networks be trained on $\mathbf{y} = \mathbf{z} + \sqrt{\varepsilon}\,\boldsymbol{\xi}$. Then $\mathbf{y} = \mathbf{x} + \boldsymbol{\eta} + \sqrt{\varepsilon}\,\boldsymbol{\xi}$, so given $\mathbf{y}$ the clean window $\mathbf{x}$ has the posterior of Section 3.3 at scale $\varepsilon + \sigma^2$, with mean $m_{\varepsilon+\sigma^2}(\mathbf{y})$ and covariance $\varepsilon_+\, 2\Gamma_{\varepsilon+\sigma^2}(\mathbf{y})$. Given $\mathbf{x}$ and $\mathbf{y}$, the two Gaussian terms split $\mathbf{y} - \mathbf{x}$ in proportion to their variances:

$$\mathbf{z} \mid \mathbf{x}, \mathbf{y} \;\sim\; N\Big(\tfrac{\varepsilon\,\mathbf{x} + \sigma^2 \mathbf{y}}{\varepsilon_+},\; \tfrac{\sigma^2 \varepsilon}{\varepsilon_+} I\Big).$$

Taking the mean over $\mathbf{x}$ gives $m_\varepsilon^{obs}(\mathbf{y}) = \mathbb{E}[\mathbf{z} \mid \mathbf{y}] = (\sigma^2 \mathbf{y} + \varepsilon\, m_{\varepsilon_+}(\mathbf{y}))/\varepsilon_+$, which is the first line of (9). The law of total covariance gives

$$\mathrm{Cov}[\mathbf{z} \mid \mathbf{y}] = \tfrac{\sigma^2\varepsilon}{\varepsilon_+} I + \tfrac{\varepsilon^2}{\varepsilon_+^2}\, \mathrm{Cov}[\mathbf{x} \mid \mathbf{y}] = \tfrac{\sigma^2\varepsilon}{\varepsilon_+} I + \tfrac{\varepsilon^2}{\varepsilon_+}\, 2\Gamma_{\varepsilon_+}(\mathbf{y}),$$

and dividing by ε gives the second line. Hence $I - 2\Gamma_\varepsilon^{obs} = \frac{\varepsilon}{\varepsilon_+}(I - 2\Gamma_{\varepsilon_+})$, and the score built from the observed moments, with both evaluated at the same point, is $(\varepsilon/\varepsilon_+)^2$ times the noise-free score at scale ε_+. The method evaluates the projector at $\hat{m}$, which moves from $m_{\varepsilon_+}(\mathbf{y})$ towards $\mathbf{y}$ by the fraction σ^2/ε_+ of the residual, so for the method the identity is approximate. We checked both identities numerically on a discrete μ (agreement to 10^{-16}).

C. The error bound and the detection margin

Where the projector matters. Consider an exact projector at $\mathbf{p}^*$ and a denoised point offset from $\mathbf{p}^*$ only tangentially, by a length t. The denoiser residual's error is then $\sqrt{d^{*2} + t^2} - d^*$, which equals t on $\mathcal{M}$ but is only about $t^2/2d^*$ when $t \ll d^*$. In this ideal case the projector matters most near $\mathcal{M}$, for healthy windows, and little for distant anomalies. On synthetic shapes its measured effect is indeed on the scores of healthy windows, largest with uneven sampling density. On the development series the median of $\|\hat{\Pi}^{\|}\mathbf{v}\|^2/\|\mathbf{v}\|^2$ is 0.07 for healthy windows and 0.06 for anomalous ones; with so small a tangent share the projector can change the ranking little, consistent with Table 1.

Rescaling. Only the ranking within a series matters, so multiplying every score by a common constant changes nothing. Applying Theorem 4.1 to the matrix $I - \hat{\Pi}^{\perp}/c$, whose score is $\hat{d}^{\perp}/c$ and whose denoiser terms are unchanged, shows that Corollary 4.2 holds with Δ_Π replaced by $\inf_{c>0} \sup \|\Pi^{\perp}_{\mathbf{p}^*} - \hat{\Pi}^{\perp}/c\|$ over the tube. A projector estimate that is correct up to a scale, such as the noisy target of (9) when $2\Gamma_{\varepsilon+\sigma^2}$ is the exact projector, then has no projector error, although the noisy denoiser's normal offset, about $\sigma^2 d^*/(\varepsilon + \sigma^2)$, remains in $\Delta_m^{\perp}$. On the development series we assume at most 16 tangent directions and take, for each series, the median over windows of the Weyl lower bound on this rescaled error at the best common c. On the 39 series with $L > 16$ its median is 0.34 (median $c = 0.82$), against 0.41 without rescaling on the same series. With $L \le 16$ the assumption does not constrain the bound, which is then uninformative.

The margin. Figure 3 illustrates Corollary 4.2 for fixed network errors. The band of possible scores widens with the distance from $\mathcal{M}$, because the projector's error acts on the whole residual.

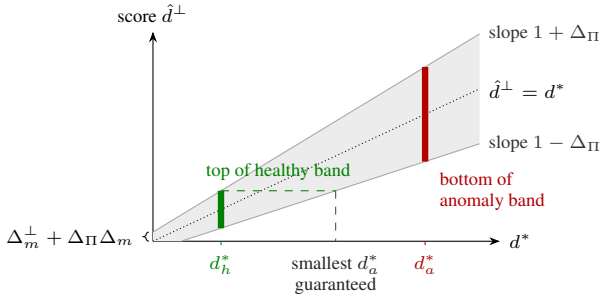


Figure 3. Corollary 4.2. Each score lies in the grey band around its true distance, which widens with d^*. An anomaly is guaranteed to outscore a healthy window once the bottom of its bar (red) lies above the top of the healthy bar (green). Drawn with $\Delta_m^{\perp} = 0.1$, $\Delta_m = 0.15$, $\Delta_{\Pi} = 0.3$.

D. Evaluation and settings

Evaluation check. The VUS sliding window of each series is the benchmark's `find_length_rank` (rank 1) on the whole raw series, and the VUS-PR is the benchmark's `generate_curve` with `thre=250` and `version='opt'`, as in the official evaluation code (TSB-AD commit `e0975a5`). Our recorded sliding windows equal an independent computation of this rule on all 48 Tuning and 350 Eva series, and the benchmark's own `get_metrics` applied to our point scores reproduces our VUS-PR to six decimals on the six Tuning series checked.

Period rule. The window length L is `find_length_rank` (rank 1) applied to the healthy prefix: the autocorrelation of the first 20 000 points is computed up to lag 400, and L is the lag (≥ 3) of its highest local maximum, or 125 if that lag exceeds 300 or no maximum exists. On Tuning, 7 of 48 series get $L = 125$.

Networks and training. Both networks are residual MLPs with four feature-wise linearly modulated blocks of width 1024, conditioned on a sinusoidal embedding of $\log \varepsilon$; the metric network outputs a rank-16 factor. Training uses AdamW (learning rate 10^{-4}, batch 1024, gradient clipping 1.0) for 3000 epochs on all prefix windows, with $\log \varepsilon_{tr} \sim N(\log \varepsilon_{eval}, 1.2^2)$ and ε_{tr} clamped to $[10^{-6}, 16]$. The denoiser is trained first and then frozen.

Seeds. The held-out run uses seed 0. On Tuning, 30 seeds give a 48-series mean VUS-PR of 0.521 with standard deviation 0.0095; seed 0 gives 0.533. On Eva, seeds 1–4 were run after the frozen run with nothing else changed: the 350-series means for seeds 0–4 are 0.594, 0.595, 0.595, 0.594 and 0.597 (standard deviation 0.0014), and each difference in Table 1 moves by at most 0.004. Single series are less stable: 9 of 350 change by more than 0.5 between seeds. Most series that vary strongly across seeds have a single labelled anomaly, so their VUS-PR depends largely on whether that anomaly's peak score ranks above every healthy peak.

Pipeline baselines. The five pipeline baselines share the prefix normalisation of the method; all except the forecast residual also share its windows and readout. When a training window is scored by a method that uses the training windows directly (nearest neighbour, closed-form CDC score), training windows within L steps of it are excluded (for the closed-form CDC score, in its first, barycentre step only). Nearest neighbour: Euclidean distance to the closest training window. Denoiser residual: $\|\mathbf{y} - \hat{m}(\mathbf{y})\|$ with the method's seed-0 denoiser at ε_{eval}. PCA residual: $\|(I - VV^{\top})(\mathbf{y} - \boldsymbol{\mu})\|$ with the mean and the leading p principal directions of the training windows, p the number of eigenvalues above $\hat{\sigma}^2(1 + \sqrt{L/N})^2$, $\hat{\sigma}^2$ the median eigenvalue and N the number of training windows. Closed-form CDC score: (5) evaluated exactly over the 8192 nearest training windows at ε_{eval}, with the projector at the closed-form barycentre. Forecast residual: the absolute one-step error of a ridge autoregression of order L fitted on the prefix ($\lambda = 10^{-3}$ times the mean diagonal of the Gram matrix), read point by point.

Cost. Training FLOPs are counted from the architecture, not profiled. Each network trains for $3000\lceil N/1024 \rceil$ steps of 1024 windows, where N is the number of prefix windows; series with at most 1024 prefix windows (108 of 350) therefore

train for 3000 steps. A network with P parameters costs about $6P$ FLOPs per training window (forward and backward), and the frozen denoiser adds $2P$ per window while the metric network trains. Scoring a window costs $2P$ for each network. The measured training times correspond to a nearly constant 19 TFLOP/s (interquartile range 18.8–19.7) across series.

Readout and published detectors. Our readout raises the VUS-PR of TSB-AD's own nearest-neighbour and PCA detectors on Tuning. Those detectors give each window's score to its centre point. We re-implemented both and changed one setting at a time towards ours. Using our Gaussian readout alone raises their VUS-PR by $+0.052$, $[+0.025, +0.084]$ (nearest neighbour), and $+0.013$, $[+0.001, +0.025]$ (PCA). The readout is the only change whose interval excludes zero for both detectors. For PCA, replacing the weighted distance to all principal components by the residual outside the noise-threshold subspace has a larger estimated effect. We have not repeated this comparison on Eva. There the published scores of these detectors are 0.350 and 0.423, and our pipeline's nearest-neighbour distance and PCA residual score 0.568 and 0.497 (Table 1). Applied as a smoothing of the point scores of the zero-shot Time-RCD checkpoint (Lan et al., 2026) on Eva, the same Gaussian lowers its VUS-PR by 0.034, $[-0.056, -0.011]$, with the largest losses on families with short anomalies (YAHOO, TODS), so the readout is not a generic gain.

Euclidean distance (Tuning). Window level and amplitude carry information that a scale-free distance discards. For the nearest-neighbour distance on Tuning, cosine distance lowers the mean VUS-PR from 0.505 to 0.374 (-0.131, $[-0.227, -0.039]$), and z-normalising each window gives 0.419 (-0.086, $[-0.202, +0.031]$). We therefore keep the Euclidean geometry of prefix-normalised windows.

Cost against pretrained models. Our two networks have 21–27M parameters together. For reference, the released Time-RCD checkpoint has 37.1M parameters[1] and is pretrained on a large synthetic corpus (Lan et al., 2026). General time-series foundation models are larger: 200M parameters for TimesFM (Das et al., 2024), 385M for MOMENT-large (Goswami et al., 2024) and 710M for the largest Chronos model (Ansari et al., 2024). These models are pretrained once and reused, whereas ours are trained for each series.

Table 2. Mean VUS-PR on Eva by family (families with at least 8 series), our method (seed 0) and TimeRCD_MAFT.

family	n	ours	TimeRCD_MAFT	ours − TimeRCD_MAFT
UCR	70	0.490	0.356	+0.134
SMD	33	0.786	0.682	+0.104
YAHOO	30	0.434	0.870	−0.436
Exathlon	30	0.972	0.969	+0.003
OPPORTUNITY	27	0.537	0.729	−0.192
NAB	23	0.500	0.430	+0.070
WSD	20	0.498	0.527	−0.029
SVDB	18	0.784	0.268	+0.515
SMAP	17	0.581	0.513	+0.068
IOPS	15	0.450	0.526	−0.075
TODS	13	0.602	0.891	−0.288
Stock	8	0.666	0.884	−0.218
NEK	8	0.772	0.824	−0.052
MGAB	8	0.578	0.049	+0.529
LTDB	8	0.686	0.496	+0.189

[1]Model size reported for https://huggingface.co/thu-sail-lab/Time-RCD.